the spongy cells contain lesser number of chloroplasts. Spongy cells facilitate the **exchange of gases** with the help of air spaces. The air space that is found next to the stomata is called **respiratory cavity or substomatal cavity**.

Vascular Tissues

Vascular tissues are present in the veins of leaf. Vascular bundles are **conjoint, Collateral and closed.** Xylem is present towards the upper epidermis, while the phloem towards the lower epidermis. Vascular bundles are surrounded by a compact layer of parenchymatous cells called **bundle sheath or border parenchyma**.

Xylem consists of metaxylem and protoxylem elements. Protoxylem is present towards the upper epidermis,while the phloem consists of sieve tubes, companion cells and phloem parenchyma. Phloem fibres are absent. Xylem consists of vessels and xylem parenchyma. Tracheids and xylem fibres are absent.

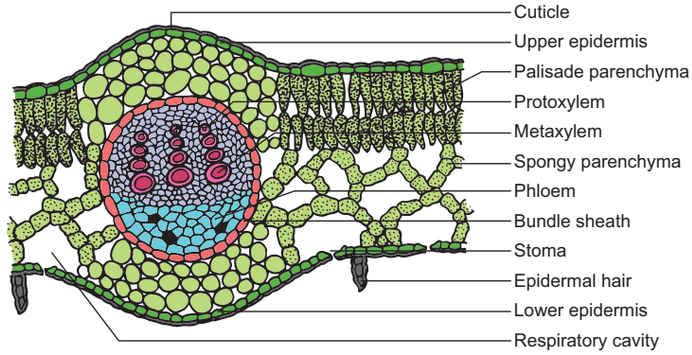


Figure 9.20: T.S. of Dicot Leaf (Sunflower)

Anatomy of a Monocot Leaf – Grass Leaf

A transverse section of a grass leaf reveals the following internal structures.

Epidermis

The leaf has upper and lower epidermis. They are made up of a single layer of thin walled cells. The outer walls are covered by thick cuticle.

The number of stomata is more or less equal on both the epidermis. The stomata is surrounded by **dumb – bell shaped** guard cells. The guard cells-contain chloroplasts, whereas the other epidermal cells do not have them.

Some special cells surround the guard cells. They are distinct from other epidermal cells.

These cells are called **subsidiary cells.**

Some cells of upper epidermis are large and thin walled. They are called **bulliform cells** or motor cells. These cells are helpful for the rolling and unrolling of the leaf according to the weather change.

Some of the epidermal cells of the grass are filled with silica. They are called **silica cells**.

Mesophyll

The ground tissue that is present between the upper and lower epidermis of the leaf is called **mesophyll.** Here, the mesophyll is not differentiated into **palisade and spongy parenchyma.** All the mesophyll cells are nearly isodiametric and thin walled. These cells are compactly arranged with limited intercellular spaces. They contain numerous chloroplasts.

Vascular Bundles

Vascular bundles differ in size. Most of the vascular bundles are smaller in size. Large bundles occur at regular intervals.Two patches of sclerenchyma are present above and below the large vascular bundles. These sclerenchyma patches give mechanical support to the leaf. The small vascular bundles do not have such sclerenchymatous patches. Vascular bundles are **conjoint, collateral and closed**. Each vascular bundle is surrounded by a parenchymatous bundle sheath. The cells of the bundle sheath generally contain starch grains. The xylem of the vascular bundle is

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